

Valid but Wrong? Untangling Validity and Soundness

An original lesson plan inspired by **IEP contributors's** the Internet Encyclopedia of Philosophy entry on Validity and Soundness. Not affiliated with or endorsed by Internet Encyclopedia of Philosophy or IEP contributors. Explore the original course: <https://iep.utm.edu/val-snd/>

GRADE LEVEL

Grades 9-12

DURATION

48 minutes

TOPICS

Logic

“Can an argument be logically perfect and still be completely wrong?”

Learning Objectives

By the end of this lesson, students will be able to:

- Define validity as a property of an argument's form: if the premises were true, the conclusion would have to be true.
- Define soundness as validity plus true premises, and explain why soundness requires more than validity alone.
- Correctly classify a set of sample arguments as valid or invalid and sound or unsound, including tricky cases where a valid argument has a false premise or conclusion.

Materials Needed

- Handout with the definitions of validity and soundness and two to three worked examples of each
- Set of at least eight sample arguments on cards, covering all four combinations of valid/invalid and true/false premises
- Whiteboard with a two-question flowchart: 'Would the conclusion have to be true if the premises were true?' then 'Are the premises actually true?'
- Sticky notes (two colors) for a class-wide sorting activity

Lesson Procedure

Hook: The Silly Valid Argument

7 min

Present this argument aloud: 'All cats are made of pizza. Garfield is a cat. Therefore, Garfield is made of pizza.' Ask students whether the conclusion follows from the premises. Most will recognize that if you accepted the premises, the conclusion would have to follow, even though it's absurd, introducing the idea that validity has nothing to do with whether the statements are actually true.

Direct Instruction: Two Separate Questions

12 min

Teach the formal definitions of validity (it is impossible for the premises to be true and the conclusion false) and soundness (a valid argument whose premises are actually true). Walk through the two-question flowchart on the board using the pizza-cat example, which is valid but unsound, and a second example that is both valid and sound.

Guided Practice: Sort the Cards

15 min

In small groups, students work through a set of sample arguments, applying the flowchart to each, first asking whether it's valid and then whether it's sound. Include at least one invalid argument with a true conclusion and true premises, showing students that valid, true-seeming, and sound are three genuinely different properties.

Whole-Class Check and Misconception Hunt

9 min

Go through two or three of the trickiest cards together as a class, focusing on the ones most groups disagreed about. Explicitly name the most common misconception: assuming a true conclusion means an argument is valid, or that a valid argument must be sound.

Closure: Exit Ticket

5 min

Students write one original two-premise argument that is valid but unsound, and one sentence explaining which premise is false and why the argument is still valid despite that.

Discussion Questions

- 1 Why does it matter to separate whether the conclusion follows from the premises from whether the premises are actually true?
- 2 Can an argument have a true conclusion and still be invalid? Can it have a false conclusion and still be valid? Explain with an example.
- 3 Why might people be more easily persuaded by an invalid argument with a conclusion they already agree with than by a valid one with a conclusion they dislike?
- 4 Is it possible to know an argument is valid without knowing whether it's sound? Is it possible to know it's sound without checking validity first?
- 5 In everyday arguments in politics, advertising, or social media, which is more common: invalid arguments, or valid-but-unsound ones? Why might that be?

Assessment

Give students six new arguments they have not seen in class, each requiring a two-part judgment of valid or invalid and, if valid, sound or unsound, with one sentence of justification per argument identifying either the logical gap or the false premise. Grade for correct application of the definitions, not for agreement with the arguments' conclusions.

Extension Activity



Have students find a real argument from a news article, advertisement, or social media post, write it out in premise/conclusion form, and evaluate it for both validity and soundness, explaining in a short paragraph whether the argument's persuasiveness depends on genuine logical force or on something else, like a false but appealing premise.

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